Selection and evolution

A Level Biology

Variation

Variation 变异 means the differences between individuals. It has three possible causes:

- **genetic** factors only —set by the **alleles** 等位基因 you inherit (for example human blood group).
- **environmental factors** 环境因素 only —set by your surroundings (for example a scar, or the language you speak).
- a **combination** of both —most features, such as height and body mass, depend on genes **and** on diet and lifestyle.

There are two patterns of variation:

- **discontinuous variation** 不连续变异—clear, separate groups with nothing in between (for example blood group A, B, AB or O). It is usually controlled by one or a few **genes** 基因, with little effect from the environment.
- **continuous variation** 连续变异—a smooth range from one extreme to the other (for example height). It is controlled by many genes together, plus the environment.

To compare the means of two samples (for example the heights of plants in sun and in shade), you use the **t-test**, which tells you whether the difference is large enough to be real, or just due to chance.

Natural selection

A **population** 种群 produces far more **offspring** 后代 than can survive, so the offspring must **compete** 竞争 for resources such as food and space. This is the "struggle for existence". The individuals that are best **adapted** 适应 are most likely to survive, **reproduce** 繁殖, and pass on their alleles to the next generation. Over many generations, the helpful alleles become more common in the population. This is **natural selection** 自然选择.

Environmental conditions can push selection in three ways:

- **stabilising selection** 稳定选择 favours the average and removes the extremes (the population stays the same).
- **directional selection** 定向选择 favours one extreme, so the mean shifts that way.
- **disruptive selection** 分裂选择 favours both extremes and removes the average.

Allele frequencies can also change in other ways:

- the **founder effect** 奠基者效应—a few individuals start a new population, so they carry only some of the alleles of the original group.
- **genetic drift** 遗传漂变—in a small population, allele frequencies change by chance from generation to generation.
- the **bottleneck effect** 瓶颈效应—a sudden fall in population size leaves few survivors, reducing the variety of alleles.

Antibiotic resistance as natural selection

A chance **mutation** 突变 makes a few bacteria **resistant** to an **antibiotic** 抗生素. When the antibiotic is used, the non-resistant bacteria die, but the resistant ones survive and reproduce. Over time the **resistance** 耐药性 spreads through the population. This is natural selection in action.

The Hardy–Weinberg principle

The **Hardy–Weinberg principle** 哈迪-温伯格原理 lets you calculate the **allele frequencies** 等位基因频率 and **genotype** 基因型 frequencies in a population. It only holds true when there is a large population, mating is random, and there is no mutation, no migration and no natural selection.

Selective breeding (artificial selection)

In **selective breeding** 选择育种, also called **artificial selection** 人工选择, humans (not nature) choose which organisms breed, so that useful features are passed on. Examples:

- breeding **disease resistance** 抗病性 into varieties of wheat and rice.
- using **inbreeding** 近交 and **hybridisation** 杂交 (crossing different lines) to make vigorous, uniform maize.
- breeding dairy cattle to improve their milk **yield** 产量.

Evolution

Evolution 进化 is the slow formation of new **species** 物种 from earlier ones, as the **gene pool** 基因库 (all the alleles in a population) changes from generation to generation.

DNA **sequence** 序列 data can show how closely related two species are: the more similar their DNA sequences, the more recently they shared a common ancestor.

Speciation 物种形成 happens when two populations become genetically separated, so they can no longer breed together. This genetic **isolation** 隔离 can come about in two ways:

- **allopatric speciation** 异域物种形成—the populations are kept apart by a **geographical separation** 地理隔离, such as a sea or a mountain range.
- **sympatric speciation** 同域物种形成—the populations live in the same area but are separated by differences in behaviour or way of life.